

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 1.3: QUADRATIC EXPRESSIONS AND EQUATIONS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Think back to the shot put thrown at the Nairobi school sports day, the tiled courtyard, and the Kericho tea farmer's shamba. Answer each question in the space below it, in full sentences, and show all your working. In Part 1 use the height rule from Lesson 1, $h = -0.1x^2 + 1.2x + 1.5$, where h is the height in metres and x is the horizontal distance in metres. In Parts 3, 4 and 5 use the simplified height rule $h = -x^2 + 6x$, which we used in later lessons to practise factorising and solving. Support every claim with evidence from the lessons, such as your calculations, your area diagrams, your Driving Question Board entries and your parabola model. Your three-part answer to the driving question comes at the end: How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems?

Part 1: Why a straight line cannot describe the throw

On sports day, your classmate's slow-motion video showed the shot put tracing a smooth curve, not a straight line. (a) Using the Lesson 1 height rule $h = -0.1x^2 + 1.2x + 1.5$, work out the height of the ball when $x = 0, 2, 4, 6, 8, 10$ and 12 metres. (b) Find the first differences (change in h from one point to the next) and then the second differences. (c) Use your results to explain why a linear equation $y = mx + c$ cannot fit this curve, and say which kind of expression is needed and what feature of it produces the rise-and-fall shape.

- (a) Substituting each value into $h = -0.1x^2 + 1.2x + 1.5$ gives: $x = 0 \rightarrow h = 1.5$; $x = 2 \rightarrow 3.5$; $x = 4 \rightarrow 4.7$; $x = 6 \rightarrow 5.1$; $x = 8 \rightarrow 4.7$; $x = 10 \rightarrow 3.5$; $x = 12 \rightarrow 1.5$. These are the 7 data points, and they plot as a smooth arch.
- (b) First differences: $+2.0, +1.2, +0.4, -0.4, -1.2, -2.0$. Second differences: $-0.8, -0.8, -0.8, -0.8, -0.8$ (constant).
- (c) A linear equation $y = mx + c$ changes by the same amount for every equal step in horizontal distance. Here the first differences are not constant: they go from $+2.0$ down to -2.0 , so the ball rises and then falls. A straight line can never do this. The constant second differences are the signature of a quadratic relationship. So we need a quadratic expression, one whose highest power is x^2 (the form $ax^2 + bx + c$ with $a \neq 0$). Here $a = -0.1$ is negative, and the $-0.1x^2$ term makes the outputs increase to a highest point and then decrease, which matches the observed rise-and-fall path of the ball.

Part 2: Forming quadratic expressions from area

Quadratic expressions arise whenever two lengths that both depend on a variable are multiplied. (a) The tiled courtyard has side lengths $(x + 2)$ and $(x + 5)$ metres. Write and expand an expression for its area using the grid/box method or FOIL, and write it in standard form $ax^2 + bx + c$. (b) Explain in one or two sentences why this area expression is quadratic and not linear. (c) The Kericho tea farmer's shamba is a rectangle with sides x and $(x + 4)$ metres. Write the area expression and expand it.	(a) Area = $(x + 2)(x + 5)$. Using FOIL: First $x \cdot x = x^2$; Outer $x \cdot 5 = 5x$; Inner $2 \cdot x = 2x$; Last $2 \cdot 5 = 10$. Collecting like terms: $x^2 + 5x + 2x + 10 = x^2 + 7x + 10$. In standard form, $a = 1$, $b = 7$, $c = 10$. (b) Both side lengths contain the variable x, so multiplying them produces an x^2 term. The highest power of x is 2, so the expression is quadratic. A linear expression has x only to the power 1. (c) Area = $x(x + 4) = x^2 + 4x$. Again an x^2 term appears because both lengths depend on x. This shows that quadratic expressions arise naturally from area as well as from the curved path of the shot put.
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Part 3: Factorising: reversing the expansion

Factorisation reverses expansion. (a) Starting from the expanded courtyard area $x^2 + 7x + 10$, factorise it and explain how you chose your two numbers. (b) The Kericho shamba's area expression is $x^2 - 81$. Factorise it and name the identity you used. (c) Factorise the simplified shot put height expression $-x^2 + 6x$. (d) Check your answer to (a) by re-expanding.	(a) I look for two numbers that multiply to $c = 10$ and add to $b = 7$. The pairs of factors of 10 are 1 and 10 (sum 11) and 2 and 5 (sum 7). So 2 and 5 work, and $x^2 + 7x + 10 = (x + 2)(x + 5)$. These are the courtyard's two side lengths. (b) $x^2 - 81 = x^2 - 9^2$, which is a difference of two squares. Using the identity $a^2 - b^2 = (a + b)(a - b)$ with $a = x$ and $b = 9$: $x^2 - 81 = (x + 9)(x - 9)$. (c) Both terms contain a common factor of x, and the leading term is negative, so I take out $-x$: $-x^2 + 6x = -x(x - 6)$. (d) Re-expanding: $(x + 2)(x + 5) = x^2 + 5x + 2x + 10 = x^2 + 7x + 10$, which is the original expression, so the factorisation is correct.
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Part 4: Solving equations to predict and decide

(a) Set the simplified shot put height rule equal to zero, $h = 0$, and solve by factorising. Explain what each solution means for the ball. (b) A Kericho tea farmer wants to fence a rectangular shamba of area 40 m^2 with one side 3 m	(a) $h = 0$ gives $-x^2 + 6x = 0$, so $-x(x - 6) = 0$. By the zero-product property, $-x = 0$ or $x - 6 = 0$, so $x = 0$ or $x = 6$. $x = 0 \text{ m}$ is the launch point where the ball is at ground level, and $x = 6 \text{ m}$ is the landing point. So we can predict that the ball lands 6 m away before it lands. (b) Area = $x(x + 3) = 40$, so $x^2 + 3x = 40$, which gives $x^2 + 3x - 40 = 0$ in standard form. I need two numbers that multiply to -40 and add to 3: these are 8 and -5. So $(x + 8)(x - 5) = 0$, giving $x = -8$ or $x = 5$. The shorter side is 5 m and the longer side is $5 + 3 = 8 \text{ m}$. The shamba is 5 m by 8 m (check: $5 \times 8 = 40 \text{ m}^2$). (c) A length cannot be negative, so $x = -8$ is mathematically a root but physically impossible, and I reject it. I keep $x = 5$. In the shot put problem both roots are meaningful: $x = 0$ is the launch and $x = 6$ is the landing. Every root must be interpreted in its real-world
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longer than the other. Let the shorter side be x metres. Form a quadratic equation in standard form, solve it by factorisation, and state the dimensions of the shamba. (c) Explain what you do with each root and why.	context.
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Part 5: Answering the driving question

Driving question: How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems, from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho? Write a three-part answer. Part A, CLAIM: state your answer in 2–3 sentences. Part B, EVIDENCE: use at least two different contexts from our lessons and include at least two pieces of calculation or data. Part C, REASONING: explain how forming, factorising and solving connect to each other and to the shot put. Finally, use the perfect square identity to rewrite the simplified height expression $-x^2 + 6x$ in a form that shows the ball's highest point, and say what it tells you about the ball's flight.	Claim: Quadratic expressions describe curved, symmetric real-life situations such as the shot put's arc and rectangular areas. Quadratic equations let us predict exact outcomes, such as where the ball lands or the dimensions of a shamba, before we measure them. Evidence: (1) Shot put at the Nairobi sports day: the height data (1.5, 3.5, 4.7, 5.1, 4.7, 3.5, 1.5 m at $x = 0, 2, 4, 6, 8, 10, 12$ m) has constant second differences of -0.8, so it is quadratic and fits $h = -0.1x^2 + 1.2x + 1.5$. In the simplified rule $h = -x^2 + 6x$, setting $h = 0$ gives $-x(x - 6) = 0$, so the ball lands at $x = 6$ m. (2) Kericho shamba: area 40 m^2 with one side 3 m longer gives $x^2 + 3x - 40 = 0$, which factorises as $(x + 8)(x - 5) = 0$, so the shamba is 5 m by 8 m, rejecting $x = -8$. (3) Tiled courtyard: $(x + 2)(x + 5)$ expands to $x^2 + 7x + 10$ and factorises back again. Reasoning: Forming a quadratic expression means multiplying two lengths that depend on x, which produces an x^2 term. Factorising reverses this by recovering the two original factors, using a common factor, a product-and-sum pair, or the difference of two squares $a^2 - b^2 = (a + b)(a - b)$. Setting the factorised equation equal to zero and applying the zero-product property gives the roots, and these roots carry meaning in context, such as landing distance or a side length. We reject roots that are impossible, such as a negative length. Perfect square form: $-x^2 + 6x = -(x^2 - 6x) = -(x^2 - 6x + 9 - 9) = -(x - 3)^2 + 9$, using $(a - b)^2 = a^2 - 2ab + b^2$ with $a = x$ and $b = 3$. Because $(x - 3)^2$ is never negative, the largest height occurs when $x = 3$, giving a maximum height of 9 m at a horizontal distance of 3 m. This is the turning point of the parabola, halfway between the launch at 0 and the landing at 6 m.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific data and results from the shot put, the Kericho shamba and the courtyard accurately, with correct	Uses evidence from at least two contexts, with minor numerical slips that do not change the conclusion.	Gives general statements with little or no specific data, or uses numbers that do not match the lessons.

	numbers throughout.		
Mathematical accuracy (expanding, factorising, solving)	All expansions, factorisations and solutions are correct, shown step by step, and checked by re-expanding or substituting.	Methods are correct with one or two small errors, such as a sign slip, and working is mostly shown.	Methods are incomplete or incorrect, for example a missing middle term, wrong factor pairs or the zero-product property not used.
Reasoning and connection to the phenomenon	Clearly explains why the x^2 term gives a curve, how factorisation reverses expansion, and how the roots predict the landing point.	Explains most of the connections but one link, such as roots to landing point, is brief or unclear.	Describes procedures without explaining why they work or how they relate to the shot put.
Interpreting solutions in context	Interprets every root in context, rejects the negative length with a clear reason, and explains the vertex as the maximum height at $x = 3$ m.	Interprets the roots correctly but gives a limited reason for rejecting a solution or only a partial vertex explanation.	Gives numerical answers without units or meaning, and does not reject impossible solutions.
Communication	The Claim–Evidence–Reasoning structure is complete, uses correct mathematical vocabulary, and is clearly organised.	The structure is mostly complete and the vocabulary is mostly correct.	The structure is missing parts and the vocabulary is rarely used correctly.

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